

HealthConnect Clinic Experience Lab: Project Summary

Angela Iseriehen | Data Analytics Track, AnalystLab Africa Experience Lab

Objective

To explore how a fictional clinic could use data and AI to reduce missed patient appointments.

Data

A dataset of about 5,000 appointment records with 18 columns. A quality check found no duplicate rows. Missing values in the reminder channel were not random: they lined up with appointments where no reminder had been sent.

Approach

1. Set up a GitHub repository and a Jupyter notebook, and documented the data quality assessment.
2. Prepared the data by converting dates to datetime fields and handling missing values.
3. Explored no-show rates across the main fields, defined five KPIs, and produced five Matplotlib visualizations.
4. Validated the findings in more depth and shared feature recommendations with a data science teammate who was building a prediction model.

Key findings

- About **48.5%** of appointments ended in a no-show.
- **Booking lead time** was the strongest single predictor, with no-show rates from **27.8% to 60.5%** across lead time bands.
- **Previous no-shows** were the second strongest: **55.4%** for patients with a history of missed visits, against **43.5%** for those without.
- The two effects **compound**. Across combinations of lead time and prior no-shows, the no-show rate ranged from **21.8% to 67.9%**.

Collaboration

The teammate's baseline model was a logistic regression with about 63% accuracy, 62% recall and a 0.68 ROC-AUC. I recommended two interaction features to capture the patterns above: lead time combined with previous no-shows, and reminder channel effectiveness by lead time band.

What this means for the clinic

- Long booking lead times and a history of missed visits identify the appointments most at risk, so reminders and follow-up can be targeted at them.
- Reminder data is incomplete by design, so any model or report built on it should treat "no reminder sent" as its own case.

Tools

Python, Jupyter, Matplotlib, GitHub